INCH-POUND

MIL-C-24643/57D
22 November 1994
SUPERSEDING
MIL-C-24643/57C
14 March 1994

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS2UW-42
(INCLUDING VARIATION LS2UWA-42 AND LS2UWS-42)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

This cable can be manufactured in two variations: LS2UW-42 (unarmored) and LS2UWA-42 (armored) and LS2UWS-42 (double overall shield).

REQUIREMENTS:

Qualification required.

Construction (watertight)

First - Copper conductor, tin coated, ASTM B 286, size 26-7, nominal diameter 0.020 inch.

Second - Separator (opaque white) may be used at manufacturer's option where required to provide free-stripping insulation.

Third - Cross-linked polyethylene insulation. Nominal wall thickness 0.010 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements. Telephone identification applied by method 3.

Fourth - Two insulated conductors twisted together with a nominal lay of 1-1/2 inches to form a pair.

Fifth - Forty-two pairs cabled together with a maximum lay of 16 times the overall diameter of the layer. Cabling sequence shall be consecutive, starting with no. 1, from the center outward. Fillers shall be used as necessary to form a firm, well-rounded assembly.

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Construction (watertight) (Continued)

Sixth - Binder tape of polyester, type G in accordance with MIL-I-631, applied helically with overlap.
Seventh- Braided shield of tin coated copper. Minimum coverage of 85 percent.

Construction variant for type LS2UW-42

Eighth - A separator tape applied helically with overlap.
Ninth - Cross-linked polyolefin jacket. Minimum average wall thickness 0.065 inch. Cable overall diameter shall be 0.750 minimum, 0.790 maximum inch.

Construction variant for type LS2UWA-42

Eighth - A separator tape applied helically with overlap.
Ninth - Cross-linked polyolefin jacket. Minimum average wall thickness 0.065 inch. Cable overall diameter shall be 0.800 minimum, 0.840 maximum inch.
Tenth - Braided metal armor.

Construction variant for type LS2UWS-42

Eighth - Spirally wrapped 0.002 inch thick polyethylene terephthalate (mylar) separator tape with 25 percent minimum overlap.
Ninth - Braided shield of tin coated copper. Minimum coverage of 85 percent.
Tenth - Binder tape applied helically with overlap.
Eleventh - Cross-linked polyolefin jacket. Minimum average wall thickness 0.065 inch. Cable overall diameter shall be 0.785 minimum, 0.825 maximum inch.

Notes: 1. Variant LS2UW-42 military part number - M24643/57-01UO.
2. Variant LS2UWA-42 military part number - M24643/57-02AO.
3. Variant LS2UWS-42 military part number - M24643/57-03UD.

EXAMINATION AND TESTS:

Basic electrical:	Requirements
Conductor Resistance - ohms/1000 feet at 25°C, maximum	45.19
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2000
Insulation resistance - Megohms/1000 feet minimum	
Conductor to conductor	500
Conductor to shield	500
Shield to shield (LS2UWS-42)	100

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EXAMINATION AND TESTS (Continued)

	Requirements
Jacket flaws	No failure
Component and shield continuity	No failure
Group A:	
Visual and dimensional	No failure
Characteristic impedance - at 1MHz, ohms	115 ±15
Watertightness - leakage, in^3, maximum	1.0
Group B:	
Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200 °C)	50
Cold bending, cable - at minus 30 ±2°C, over a 9 inch diameter mandrel	No damage
Drip - 95 ±1°C	Zero
Bending endurance - at 25 ±2°C, cycles minimum	25
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Physicals (unaged)	
Insulation	
Tensile strength - lb/in^2, minimum	1800
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Group C:	
Physicals - (aged)	
Insulation	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged minimum	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged minimum	60
Hot oil immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Shrinkage	No failure
Heat distortion - percent of unaged, maximum	30

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EXAMINATION AND TESTS (Continued):	Requirements
* Permanence of printing (jacket) - cycles, minimum....	125
Cable sealant removability.....	No failure
Armor - conformance to material, construction and	
* coverage (type LS2UWA - 42 only).....	No failure
Shield conformance to material, construction and coverage.....	No failure
Surface transfer impedance	
Milliohms per meter, maximum.....	700
Emp response time - db, minimum.....	60
Group D:	
Flame propagation (cable).....	No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

Aging and compatibility (cable) (125 ±5°C.)..... No failure	
Abrasion resistance (jacket) - scrapes, minimum.....	75
Acid gas equivalent - percent, maximum	
Jacket.....	2
Fillers.....	2
Insulation.....	18
Halogen content - percent, maximum	
Jacket.....	0.2
Fillers.....	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum..	50
Elongation - percent of unaged, minimum.....	50
Smoke index, maximum	
Jacket.....	25
Fillers.....	45
Insulation.....	45
Toxicity index, maximum	
Jacket.....	5
Fillers.....	5
Insulation.....	1.5
Durometer (jacket) - (type A) hardness, minimum.....	80
Weathering (jacket).....	No failure

UNIT ORDERING LENGTHS: 1000 feet nominal

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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